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Hybrid SARIMA-recurrent neural network approach for road accident forecasting

Igor Gavrilov[1*], Pavel Hrubes[1] and Emil Pelikan[2]

Abstract

This study investigates the forecasting of daily road traffic accidents using time series models, through a controlled comparative evaluation of baseline SARIMA and Hybrid SARIMA-LTSM frameworks. We employ a Seasonal Autoregressive Integrated Moving Average (SARIMA) model to capture linear and seasonal patterns in accident data from 2010 to 2023. To address potential non-linearities, we extend the approach by modelling SARIMA residuals using Recurrent Neural Networks (RNNs), including bidirectional RNN (BiRNN) and Long Short-Term Memory (LSTM) variants, to develop a hybrid SARIMA-RNN framework. The hybrid model integrates SARIMA forecasts with RNN-based residual predictions as a methodological extension whose empirical performance is rigorously evaluated against the baseline. Within residual-domain model selection, the unidirectional LSTM achieved the lowest RMSE (0.184) and MAPE (1.0966%) among tested architectures, confirming its effectiveness in capturing temporal dependencies in the log-residual space. A hybrid SARIMA-LSTM model was also evaluated to enhance residual prediction; however, the one-step-ahead rolling-origin strategy with a 7-day sliding window did not improve accuracy relative to the baseline SARIMA, yielding only minor increases in RMSE and MAPE.

Keywords Road accidents, Time series forecasting, SARIMA, Recurrent Neural Networks (RNN), Hybrid models, BiRNN, BiLSTM, Transport safety, Residual modelling

1 Introduction

Road accidents are a global issue. According to the World Health Organisation, road traffic deaths in the European Region decreased by 13% in 2020 due to government-imposed COVID-19 restrictions [1]. While this temporary decrease highlights the influence of external factors on accident rates, long-term prevention requires proactive strategies supported by accurate forecasting methods. Predictive modelling can play a critical role in helping regional authorities identify high-risk periods and locations, optimise resource allocation, and implement targeted safety measures.

In the Czech Republic, a notable downward trend in traffic accidents was observed in the post-pandemic period [2]. To better understand these dynamics, a time series analysis was conducted using the Seasonal Autoregressive Integrated Moving Average (SARIMA) model. This approach enabled the incorporation of both seasonal fluctuations and long-term trends, offering valuable insights into accident patterns. The model's findings can be used to support evidence-based policymaking aimed at improving road safety and reducing future accident rates. However, using SARIMA, introduced in the 1970s by Box and Jenkins [3], doesn't explicitly model the residual errors in prediction. The first implementation of SARIMA models for time-series prediction was in the late 1970s [4], and it remains a valuable method. Many

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studies have used the SARIMA model for road accident prediction [5, 6] as the best choice. While SARIMA models are effective at capturing linear dependencies and seasonal patterns, real-world time series often exhibit complex, non-linear dynamics that traditional statistical models might not fully capture. This can result in the residuals of a SARIMA model still containing valuable, but unmodeled, information or non-linear patterns. Artificial Neural Networks showed better results than multiple regression models in road accident prediction [7]. Also, existing comparative studies of different types of ANN for road accident predictions [8, 9]. Recurrent Neural Networks (RNNs) [10] and their specialised variants, LSTMs [11] and BRNNs, are well-suited for modelling sequential, potentially non-linear patterns in time-series data.

The remainder of the paper is organised as follows: Sect. 2, data description. Section 3: SARIMA modelling and creating a residuals dataset. Section 4 trains different types of recurrent neural networks and selects the best one based on metrics as a candidate for the hybrid SARIMA model. Section 5 compares the results of one-step-ahead prediction using SARIMA and the SARIMA hybrid version. Section 6 discussion. Section 7 conclusion.

2 Data

The dataset of road accidents [11] consists of 1,114,777 incidents reported to the Czech Republic Police between January 2010 and December 2023. According to the Czech Government law [12]. A road accident must be reported to the police if it involves (1) a fatality, or (2) property damage exceeding 4,000 EUR. Therefore, the dataset represents a sample of the total population of road accidents during this period.

The downward trend after period 2020 and 2021 is likely a post-pandemic effect, as described by the author [2], and may be attributed to reduced population mobility nationwide. This reduction was influenced by the widespread adoption of remote work by many companies and the shift to distance learning supported by universities.

For the purposes of this study, the data were aggregated by day, resulting in a time series of 5,113 daily observations. This level of granularity is well-suited for seasonal modelling and helps reduce noise commonly present in hourly counts. Daily aggregation enables the detection of long-term trends and recurring seasonal patterns, which are essential for effective forecasting. This time series is the basis for both the SARIMA modelling and the subsequent neural network analysis of the residuals.

The initial dataset was divided into two parts: (1) a training and model development period from January 2010 to December 2022, used to create SARIMA and artificial neural network models [13], and (2) an implementation and evaluation period from January 2023 to December 2023.

3 SARIMA modelling

To develop the SARIMA model and mitigate heteroscedasticity in the daily road accident counts, a natural logarithm transformation was applied to the data. This logarithmic transformation stabilises the variance and reduces the influence of extreme values, thereby enhancing the suitability of the time series for linear modelling techniques. Such preprocessing is crucial for improving model accuracy and ensuring the assumptions of stationarity and homoscedasticity are satisfied.

The preprocessing procedure applied to the accident time series is presented in Fig. 1, including the original data, log-transformed series, seasonal differencing, and second-order differencing.

To test for stationarity, both the Kwiatkowski-Phillips-Schmidt-Shin [14] and Augmented Dickey-Fuller [15] tests were applied. The original series was found to be non-stationary: the KPSS test statistic is 25.306 (p-value = 0.01), and the ADF test statistic is -9.855 (p-value = 0.01). After applying seasonal differencing with a 7-day lag, the series became approximately stationary, with a KPSS of 0.023 (p-value = 0.01) and an ADF statistic of -20.141(p-value < 0.01). Nevertheless, a second-order difference was applied as a precaution to remove any remaining minor trends and ensure that the input to the SARIMA and hybrid models met strict stationarity assumptions.

Inspection of the autocorrelation function (ACF) and partial autocorrelation function (PACF) are shown in Fig. 2, following seasonal differencing at lag 7 and second-order differencing indicates several notable features. The ACF displays spikes at lag 1 and lag 2 and a distinct seasonal spike at lag 7. The PACF demonstrates exponential decay across the seasonal lags (7, 14, 21, ...), together with gradually diminishing spikes. These patterns are consistent with the presence of both short-term autoregressive and moving-average components, as well as seasonal dependence.

On the basis of these diagnostics, the set of candidate SARIMA models is characterised by parameters $p = 4$, $d = 1$, $q \in \{0,1,2,3\}$, $P \in \{1,2\}$, $D = 1$, $Q = 1$.

3.1 Model selection strategy

For selecting the best-fitting model among the candidate SARIMA specifications, several statistical criteria and diagnostic tests were employed.

First, the Ljung-Box [16] test was applied to the residuals of each fitted model to assess the absence of autocorrelation, which indicates whether the residuals behave like white noise (Eq. 1). A non-significant Ljung-Box

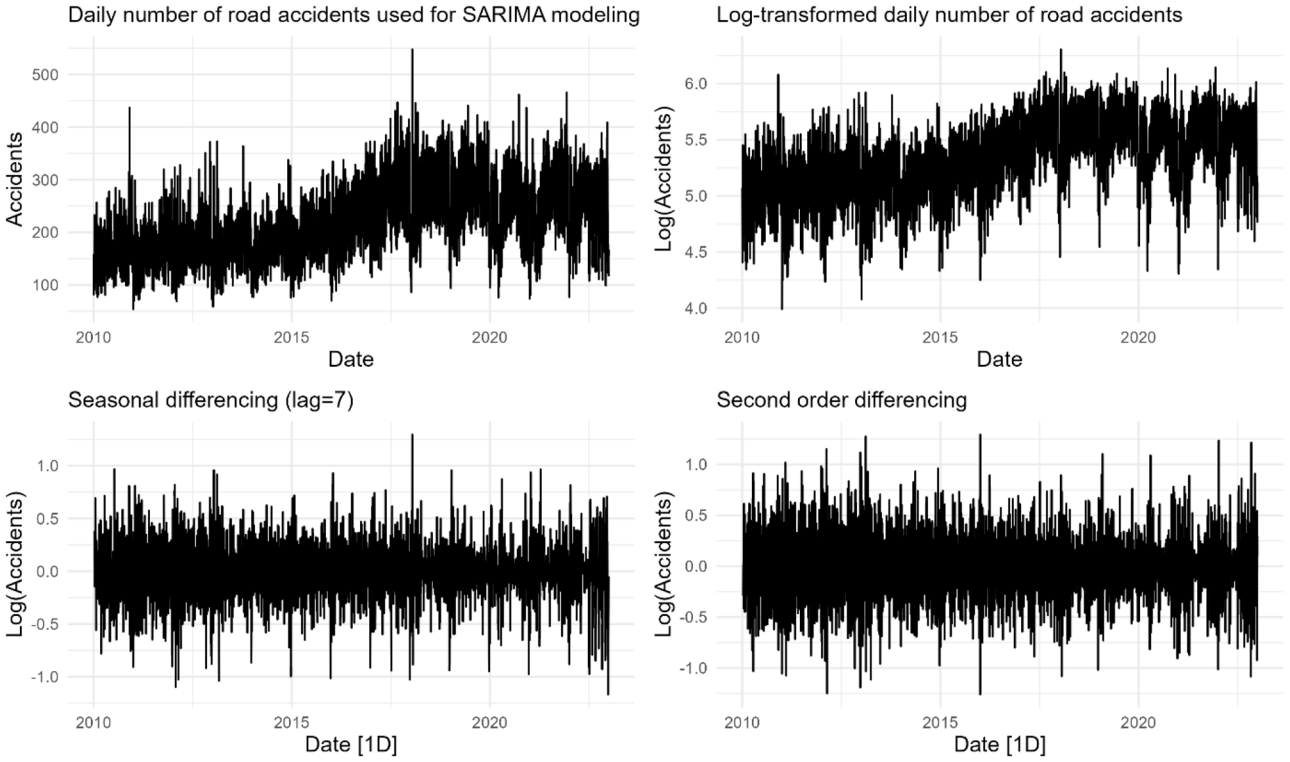


Fig. 1 Time series of daily road accident counts in the Czech Republic from 01.01.2010 to 31.12.2022 at different preprocessing stages: (upper-right) original data, (upper-left) log-transformed data, (down-right) after 7-day seasonal differencing, and (down-left) after second-order differencing

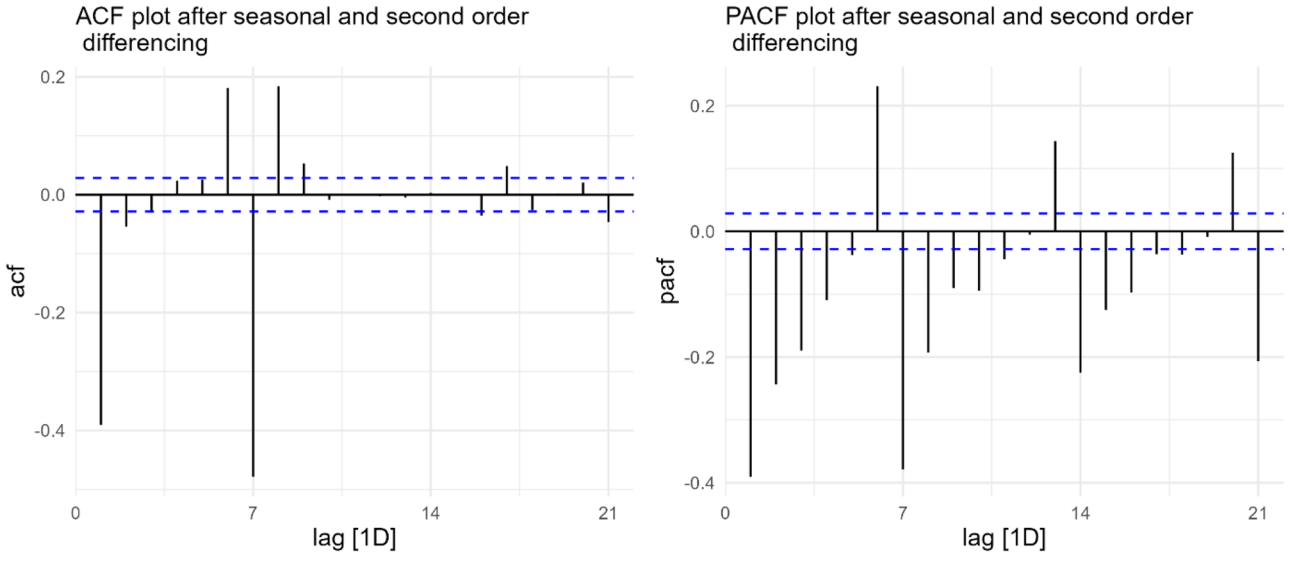


Fig. 2 Autoregressive correlation function after seasonal (lag = 7) and second order (lag = 1) differencing plot (left) and Partial autoregressive correlation function after seasonal (lag = 7) and second order (lag = 1) differencing plot (right)

test suggests that the model adequately captures temporal dependencies in the data. The Ljung-Box test was applied with h = 14 (the number of lags considered), and the degrees of freedom adjusted based on the number of parameters (p + q+P + Q) in the selected model.

$$Q^* = n(n+2)\sum_{k=1}^{h}(n-k)^{-1}r_k^2 \quad (1)$$

Second, the Box-Pierce [17] test was also used with the same lag size h = 14 as an alternative test for residual autocorrelation (Eq. 2). Similar to the Ljung-Box test, it

evaluates whether the residuals approximate white noise, providing further diagnostic confirmation.

$$Q^* = n \sum_{k=1}^{h} r_k^2 \tag{2}$$

Third, the Akaike Information Criterion (AIC) [18] was used to compare model parsimony and goodness of fit (Eq. 3). Models with lower AIC values are generally preferred, as they balance model complexity and explanatory power.

$$AIC = -2log\, L + 2m \tag{3}$$

To evaluate forecast accuracy, each candidate model was used to produce a single-step forecast horizon of length implementation dataset from 01.01.2023 to 31.12.2023 (h = 365). The quality of these forecasts was measured using two standard metrics:

- The Root Mean Squared Error (RMSE), which quantifies the average magnitude of prediction errors in the same units as the data.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2} \tag{4}$$

- The Mean Absolute Percentage Error (MAPE), which expresses forecast errors as a percentage of observed values, allowing for scale-independent comparison across models.

$$MAPE = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{y_i - \widehat{y}_i}{y_i} \right| \times 100 \tag{5}$$

The fitted SARIMA models and their corresponding evaluation metrics are summarised in Table 1. The table presents the model specifications alongside key diagnostic and performance statistics used for model selection and assessment. These include the Ljung-Box and Box-Pierce test statistics (with lag = 14), which evaluate the residuals for autocorrelation to confirm the adequacy of the model fit. Additionally, information criteria such as the Akaike Information Criterion (AIC) provide a measure of model parsimony, balancing fit quality and complexity. Single-step forecast accuracy metrics: Root Mean Squared Error (RMSE) (Eq. 4) and Mean Absolute Percentage Error (MAPE) (Eq. 5) - were calculated on back-transformed predicted data, comparing with observed data from the 2023 one-year period. These metrics are also reported to quantify each model's prediction accuracy. Collectively, these metrics enable a comprehensive comparison of candidate models, facilitating the selection of the optimal SARIMA specification for forecasting the daily number of road accident counts.

3.2 Final SARIMA model

Among all candidate models, Model 3 $SARIMA(4,1,2)(1,1,1)_7$ was selected as the optimal specification based on multiple criteria. It demonstrated the lowest AIC value (-2710.2031), indicating the best trade-off between model complexity and fit. Furthermore, both the Ljung-Box and Box-Pierce test statistics showed non-significant p-values (0.5096 and 0.5113, respectively), suggesting that the residuals approximate white noise and no autocorrelation remains unaccounted for. In terms of predictive performance, Model 3 also achieved RMSE (81.5317) and MAPE (33.2427%) for the period from 01.01.2023 to 31.12.2023. These combined diagnostics and forecasting metrics justify the selection of this SARIMA model for the time series of daily road accident counts. Model 3 $SARIMA(4,1,2)(1,1,1)_7$ has coefficients:
$\phi_1 = 0.9546$, $\phi_2 = -0.0889$, $\phi_3 = -0.0110$, $\phi_4 = 0.0326$, $\Phi_1 = 0.0659$, $\theta_1 = -1.5821$, $\theta_2 = 0.5860$, $\Theta_1 = -0.9952$, and can be represented as backshift notation:

Table 1 Summary of fitted SARIMA models with diagnostic test statistics, information criteria, and single-step forecast accuracy metrics (RMSE, MAPE) for daily road accident counts from 01.01.2023 to 31.12.2023. Residual autocorrelation was assessed using the Ljung-Box and Box-Pierce tests; p-values > 0.05

n	Model	AIC	Ljung-Box	Box-Pierce	RMSE	MAPE
1	$SARIMA(4,1,0)(1,1,1)_7$	-2441.4533	0	0	92.182	39.6554
2	$SARIMA(4,1,1)(1,1,1)_7$	-2696.3373	0.0055	0.0056	88.9021	32.1054
3	$SARIMA(4,1,2)(1,1,1)_7$	-2710.2031	0.5096	0.5113	81.5317	33.2427
4	$SARIMA(4,1,3)(1,1,1)_7$	-2706.1735	0.2027	0.2028	96.496	33.2178
5	$SARIMA(4,1,0)(2,1,1)_7$	-2439.4694	0	0	92.1124	39.6219
6	$SARIMA(4,1,1)(2,1,1)_7$	-2694.6025	0.0035	0.0035	88.3392	32.0895
7	$SARIMA(4,1,2)(2,1,1)_7$	-2708.4547	0.4459	0.4474	81.7841	33.7775
8	$SARIMA(4,1,3)(2,1,1)_7$	-2706.1835	0.2924	0.2936	82.4569	32.3219

$$\begin{aligned}&\left(1-0.9546B+0.0889B^2+0.0110B^3-0.0326B^4\right)\\&\left(1-0.0659B^7\right)(1-B)\left(1-B^7\right)\\&Y_t=(1-1.521B+0.5860B^2)(1-0.9952B^7)e_t\end{aligned} \tag{6}$$

4 Neural network models for residual prediction

4.1 Residual extraction and data preparation

Residuals from the model $SARIMA(4,1,2)(1,1,1)_7$ were divided into train, validation, and test datasets. The datasets have lengths of 3792, 474, and 474 values for the training, validation, and test sets, respectively. The residuals appear to be randomly distributed around zero, with no visible patterns, indicating that the model captures the main temporal structure of the data.

Given that the time series shows seasonal behaviour, the sequence length was set to 7, corresponding to one full seasonal cycle (e.g., days in a week). This choice allows the model to capture seasonal dependencies effectively.

The Fig. 3 shows two types of layers: one-directional and bi-directional, with a length of the input sequence of 7 steps. Input sequences were constructed using a sliding window with a step size of 1, where each prediction at time t + 1 uses only residual values from t-6 to t. No observations beyond time t were available to the model, ensuring no information leakage.

The batch size was set to 64 to balance computational efficiency and gradient estimation stability during training. A total of 40 training epochs was selected to ensure the neural network had sufficient time to learn temporal dependencies in the residual series. Preliminary experiments indicated that the validation loss consistently stabilised before 40 epochs, suggesting model convergence without overfitting. This choice balanced training efficiency with model accuracy, providing enough iterations for reliable learning while avoiding unnecessary computational cost.

4.2 Network architectures evaluated

Two types of neural networks were used to create the models: Simple Recurrent Neural Network [19] and Long-Short Term Memory Neural Network [20]. The use of LSTM models eliminates the vanishing gradient problem present in simple RNN models. Additionally, for each type of neural network, a bidirectional [21] variant was used, which allows the model to process information in both forward and backward directions.

Figure 4 shows an unfolded one-layer forward-directional Simple Recurrent Neural Network, where:

The sequence length: T.
Time-step index: t = 1, …, T.
Input vector at time t: x_t

Activation function (hyperbolic tangent):

$$\tanh(x)=\frac{(e^x-e^{-x})}{(e^x+e^{-x})} \tag{7}$$

Initial hidden state vector: h_0

Hidden state vector at time t:

$$h_t=\tanh(W_x x_t+W_h h_{t-1}+b) \tag{8}$$

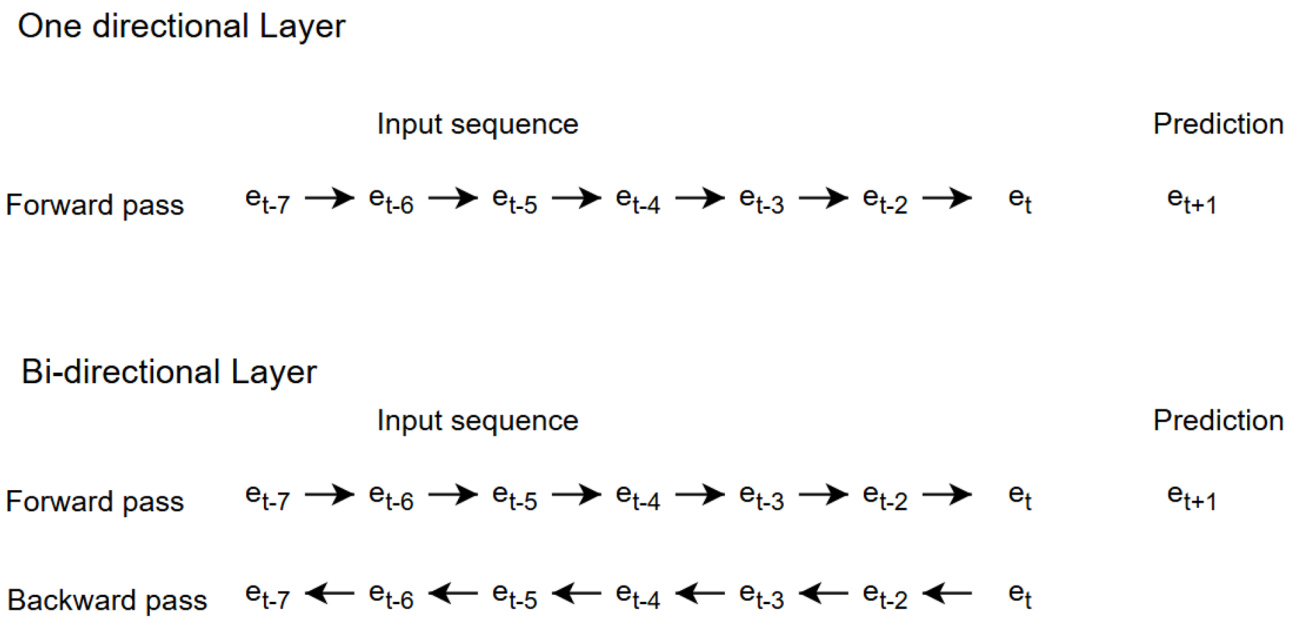


Fig. 3 One and bi-directional layers with an input sequence length of 7 and a prediction

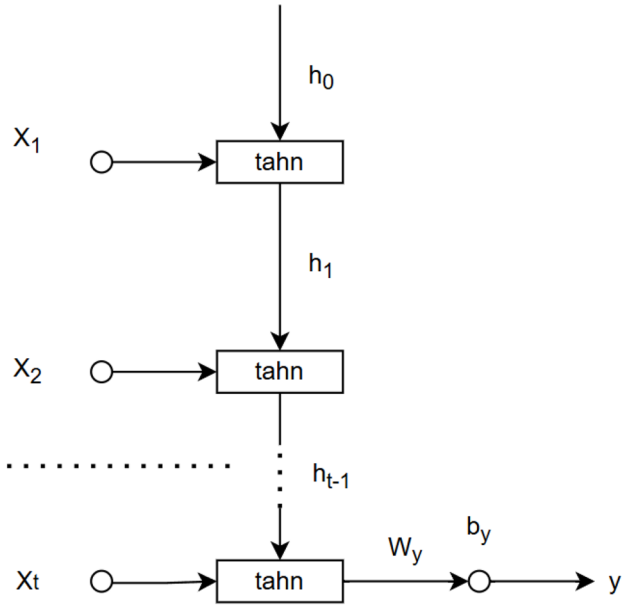


Fig. 4 Unfolded a one-layer forward-directional Simple Recurrent Neural Network (RNN). The diagram shows the network unrolled over successive time steps, with recurrent connections carrying the hidden state forward in time

- Bias vector: b
- Input weight matrix: W_x
- Recurrent weights matrix: W_h

Output vector at time t:

$$y_t = W_y h_t + b_y \tag{9}$$

- Output weights vector: W_y
- Output bias vector: b_y

As mentioned before, Simple Recurrent Neural Networks have the gradient vanishing problem, and LSTM eliminates it, as shown on Fig. 5.

The sequence length: T
Time-step index: t = 1, ..., T
Input vector at time t: x_t
Initial hidden state vector: h_0
Initial cell state vector: C_0

Activation function (hyperbolic tangent):

$$\tanh(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})} \tag{10}$$

Activation function (sigmoid):

$$\sigma(x) = \frac{1}{(1 + e^{-x})} \tag{11}$$

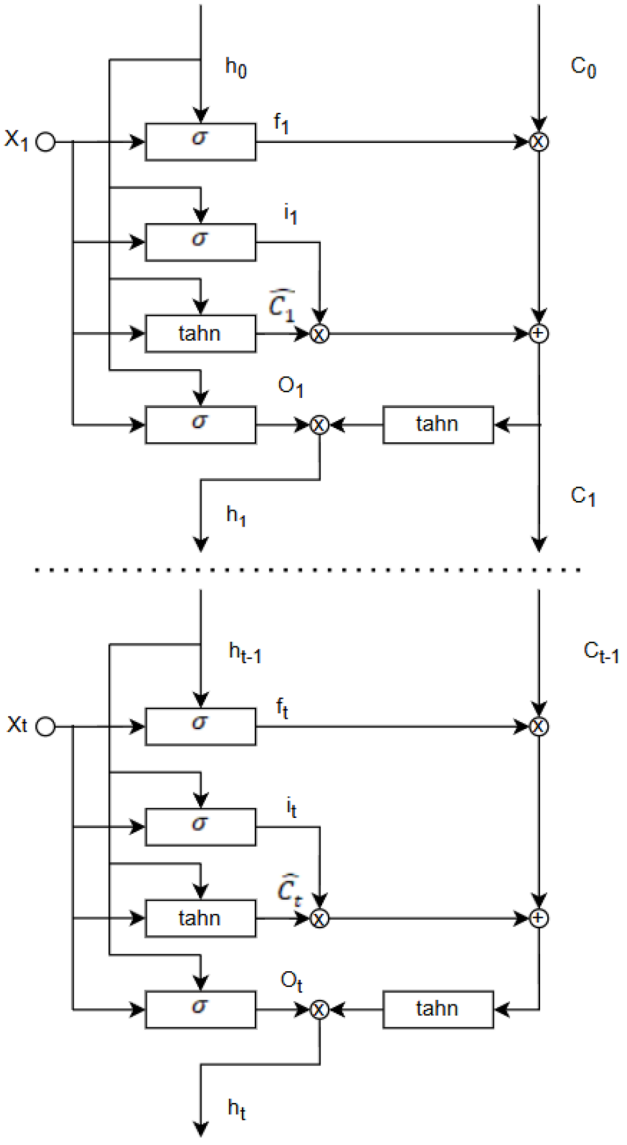


Fig. 5 Unfolded a one-layer Long Short-Term Memory (LSTM) neural network. The diagram illustrates the network unrolled over time steps, showing the flow of information through the input, forget, and output gates, as well as the cell state and hidden state

Forget gate at time t:

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{12}$$

- Input weight vector of forget gate: W_f
- Recurrent matrix weight of forget gate: U_f
- Bias vector of forget gate: b_f

Input gate at time t:

$$i_t = \sigma\left(W_i x_t + U_i h_{t-1} + b_i\right) \tag{13}$$

- Input weight vector of the input gate: W_i

- Recurrent matrix weight of input gate: U_i
- Bias vector of input gate: b_i

Candidate cell state at time t:

$$\widehat{C_t} = \tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{14}$$

- Input weight vector of cell state: W_c
- Recurrent matrix weight of cell state: U_c
- Bias vector of cell state: b_c

Cell state at time t:

$$C_t = f_t \odot C_{t-1} + \widehat{C_t} \odot i_t \tag{15}$$

Output gate at time t:

$$O_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{16}$$

- Input weight vector of output gate: W_o
- Recurrent matrix weight of output gate: U_o
- Bias vector of output gate: b_o

Hidden state vector at time t:

$$h_t = O_t \odot \tanh(C_t) \tag{17}$$

For both Simple Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks, the hidden state vector h_t at each time step serves as the output of the current layer. This sequence of hidden states can be passed as input to subsequent layers, either additional recurrent layers or dense (fully connected) layers, allowing the construction of deeper and more complex network architectures. Such stacking enables the model to learn higher-level temporal features and generate new output sequences suited for various prediction tasks.

In addition to the standard forward-directional architecture, both Simple RNN and LSTM models were extended to include bidirectional variants. Bidirectional networks [21] process the input sequence in both forward and backward directions, maintaining two separate hidden states at each time step. This approach allows the model to capture context from past and future time steps simultaneously, often improving performance on sequence prediction tasks where the full temporal context is important.

Table 2 The simple recurrent neural network model consisted of three recurrent layers followed by a dense output layer. The layer configurations and parameters are as follows:

Layer (type)	Output Shape	Number of parameters
RNN	128	16,640
Dropout (0.1)		
RNN	16	2,320
RNN	128	18,560
Dropout (0.1)		
Dense	1	129

Total parameters: 37,649

4.3 Model training and evaluation

For training, the models were implemented using the Keras [22] deep learning framework. Hyperparameter tuning was conducted to optimise model performance by varying the network architecture and regularisation parameters. Specifically, models were tested with up to three stacked recurrent layers. The number of neurons in each layer ([16,32,64,128]), dropout rates ([0.0, 0.1, 0.2]), and learning rates ([0.01,0.001, 0.0001]). As an optimiser for training models, Adaptive Movement Estimation (Adam) [23] was chosen. The search objective was the minimum validation mean absolute error, execution per epoch to ensure a robust solution. This grid search approach enabled the selection of the optimal combination of layer depth, neuron count, and dropout rate, thereby balancing model complexity and robustness during training.

Model performance during training was evaluated using Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) on the validation dataset.

Finally, the best-performing models from each category, Simple RNN, Bidirectional RNN, LSTM, and Bidirectional LSTM, were selected for testing and comparison. The detailed architecture and performance metrics are summarised below.

The structure and parameter configuration of the Simple Recurrent Neural Network model used in the experiments are provided in Table 2.

The first layer has an input-to-hidden weight matrix $W_x^{(1)} \in \mathbb{R}^{128\times 1}$, a hidden-to-hidden weight matrix $W_h^{(1)} \in \mathbb{R}^{128\times 128}$, and a bias vector $b^{(1)} \in \mathbb{R}^{128\times 1}$, totalling 16,640 parameters. The dropout doesn't introduce parameters. The second layer takes as input the output sequence from the first layer (length 128), and has the input weight matrix $W_x^{(2)} \in \mathbb{R}^{16\times 128}$, hidden-to-hidden weight matrix $W_h^{(2)} \in \mathbb{R}^{16\times 16}$, a bias vector $b^{(2)} \in \mathbb{R}^{16\times 1}$, totalling 2,320 parameters. The third layer receives the output vector from the second layer, and has an input weight matrix $W_x^{(3)} \in \mathbb{R}^{128\times 16}$, hidden-to-hidden weight matrix $W_h^{(3)} \in \mathbb{R}^{128\times 128}$, a bias vector $b^{(3)} \in \mathbb{R}^{128\times 1}$. Number of parameters: 18,560. The final dense layer maps the hidden vector h_t to a scalar output with a weight matrix $W_y \in \mathbb{R}^{1\times 128}$ and a bias $b_y \in \mathbb{R}^{1\times 1}$, which has 129 parameters. The full network contains 37,649 trainable parameters.

Table 3 Bidirectional Recurrent Neural Network model architecture comprised two bidirectional recurrent layers and a dropout layer with parameter 0.2, followed by a dense output layer:

Layer (type)	Output Shape	Number of parameters
Bidirectional RNN		
- Forward pass	128	16,640
- Backward pass	128	16,640
Bidirectional Output (concatenated)	256	33,280
Dropout (0.2)		
Bidirectional RNN		
- Forward pass	32	9,248
- Backward pass	32	9,248
Bidirectional Output (concatenated)	64	18,496
Dense	1	65

Total parameters: 54,841

Table 4 Long Short-Term Memory Neural Network consisted of an LSTM layer with one dropout layer with a parameter of 0.2, followed by a dense layer:

Layer (type)	Output Shape	Number of parameters
LSTM	32	4,352
Dropout (0.2)		
Dense	1	33

Total parameters: 4,385

The structure and parameter configuration of the Bidirectional Simple Recurrent Neural Network model used in the experiments are provided in Table 3.

The bi-directional layer has two directions: the forward pass has an input-to-hidden weight matrix $W_x^{(1,F)} \in \mathbb{R}^{128\times 1}$, hidden-to-hidden $W_h^{(1,F)} \in \mathbb{R}^{128\times 128}$, and bias vector $b^{(1,F)} \in \mathbb{R}^{128\times 1}$, that gives 16,640 parameters, similar to the backward pass, which has an input-to-hidden weight matrix $W_x^{(1,B)} \in \mathbb{R}^{128\times 1}$, hidden-to-hidden $W_h^{(1,B)} \in \mathbb{R}^{128\times 128}$, and bias vector $b^{(1,B)} \in \mathbb{R}^{128\times 1}$, which gives 16,640 parameters; total, the first layer has 33,280 learning parameters. The second layer, as an input, uses a stacked output vector with dimension $\mathbb{R}^{256\times 1}$ from the first layer and passed through forward and backwards. The forward pass: input-to-hidden weight matrix $W_x^{(2,F)} \in \mathbb{R}^{32\times 256}$, hidden-to-hidden weight matrix $W_h^{(2,F)} \in \mathbb{R}^{32\times 32}$, and a bias vector $b^{(2,F)} \in \mathbb{R}^{32\times 1}$ with 9,248 training parameters. The backward pass is similar to the forward pass, but processing backward through steps, and the input-to-hidden weight matrix $W_x^{(2,B)} \in \mathbb{R}^{32\times 256}$, hidden-to-hidden weight matrix $W_h^{(2,B)} \in \mathbb{R}^{32\times 32}$, and a bias vector $b^{(2,B)} \in \mathbb{R}^{32\times 1}$. With 9,248 training parameters, the total number of training parameters is the sum of forward and backward parameters and equals 18,496.

Table 5 The Bidirectional Long Short-Term Memory Neural Network model was composed of a bidirectional LSTM layer with one dropout layer with a parameter of 0.2, followed by a dense layer:

Layer (type)	Output Shape	Number of parameters
Bidirectional LSTM		
- Forward pass	16	1,152
- Backward pass	16	1,152
Bidirectional Output (concatenated)	32	2,304
Dropout (0.2)		
Dense	1	33

Total parameters: 2,337

The final layer contains a single output unit, and the weight matrix $W_y \in \mathbb{R}^{1\times 64}$ and a bias $b_y \in \mathbb{R}^{1\times 1}$, which has 65 parameters. The full network contains 51,841 trainable parameters. Bidirectionality doubles the parameter count because forward and backward directions maintain independent recurrent weight matrices.

The structure and parameter configuration of the Long Short-Term Memory Neural Network model used in the experiments are provided in Table 4.

The model includes a first long short-term memory layer with 32 hidden units. Each gate comprises an input weight matrix, a recurrent weight matrix, and a bias vector. For the forget gate, the input weight matrix $W_f \in \mathbb{R}^{32\times 1}$, recurrent weight matrix $U_f \in \mathbb{R}^{32\times 32}$, and bias vector $b_f \in \mathbb{R}^{32\times 1}$result in 1,088 trainable parameters. The input gate contains $W_i \in \mathbb{R}^{32\times 1}$, $U_i \in \mathbb{R}^{32\times 32}$, and $b_i \in \mathbb{R}^{32\times 1}$, totalling 1,088 parameters. The candidate cell includes $W_c \in \mathbb{R}^{32\times 1}$, $U_c \in \mathbb{R}^{32\times 32}$, and $b_c \in \mathbb{R}^{32\times 1}$, also totalling 1,088 parameters. Similarly, the output gate consists of $W_o \in \mathbb{R}^{32\times 1}$, $U_o \in \mathbb{R}^{32\times 32}$, and $b_o \in \mathbb{R}^{32\times 1}$, contributing 1,088 parameters. The dense output layer maps the hidden vector h_t to a scalar output using a weight matrix $W_y \in \mathbb{R}^{1\times 32}$ and a bias $b_y \in \mathbb{R}^{1\times 1}$, totalling 33 parameters.

The full model contains 4,385 trainable parameters. Each gate maintains independent input and recurrent weight matrices, which explains the fourfold parameter structure of the LSTM layer.

The structure and parameter configuration of the Bidirectional Long Short-Term Memory Neural Network model used in the experiments are provided in Table 5.

The model includes a bidirectional long short-term memory layer with 16 hidden units in each direction, producing a stacked output vector of dimension 32×1 for each time step. Each direction maintains independent gate parameters. For the forward pass, the forget gate

contains an input weight matrix $W_f^{(F)} \in \mathbb{R}^{16\times 1}$, a recurrent weight matrix $U_f^{(F)} \in \mathbb{R}^{16\times 16}$, and a bias vector $b_f^{(F)} \in \mathbb{R}^{16\times 1}$, resulting in 288 training parameters; the input gate contains $W_i^{(F)} \in \mathbb{R}^{16\times 1}$, $U_i^{(F)} \in \mathbb{R}^{16\times 16}$, and $b_i^{(F)} \in \mathbb{R}^{16\times 1}$, totalling 288 parameters; the candidate cell contains $W_c^{(F)} \in \mathbb{R}^{16\times 1}$, $U_c^{(F)} \in \mathbb{R}^{16\times 16}$, and $b_c^{(F)} \in \mathbb{R}^{16\times 1}$, totalling 288 parameters, and the output gate contains $W_o^{(F)} \in \mathbb{R}^{16\times 1}$, $U_o^{(F)} \in \mathbb{R}^{16\times 16}$, and $b_o^{(F)} \in \mathbb{R}^{16\times 1}$, totalling 288 parameters. The backward pass has an identical structure and parameter count. Consequently, each direction contains 1,152 parameters, and the full bidirectional LSTM layer contains 2,304 training parameters. The dense output layer maps the stacked hidden vector $h_t \in \mathbb{R}^{32\times 1}$ to a scalar using a weight matrix $W_y \in \mathbb{R}^{1\times 32}$ and a bias $b_y \in \mathbb{R}^{1\times 1}$, totalling 33 parameters. The full model contains 2,337 training parameters.

All models were subsequently evaluated on the test dataset. The Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) metrics were calculated for each neural network model based on the test dataset predictions. These metrics, presented in Table 6, provide further quantitative evaluation of model accuracy. Consistent with the residual analysis, the LSTM model achieved the lowest RMSE (0.184) and MAPE (1.0966) within residual-domain model selection (Table 6); this superiority is confined to the log-residual space and does not extend to the final accident-count forecast (Table 7). The BiRNN and BiLSTM models also showed competitive error rates, while the RNN exhibited higher errors, confirming the relative effectiveness of each architecture in this forecasting task.

Based on the implementation-phase metrics, the LSTM model, with only one layer and the lowest performance, was chosen.

Table 6 Test dataset performance metrics (RMSE and MAPE) for neural network models

Model type	RMSE	MAPE
RNN	0.1888	1.9416
Bidirectional RNN	0.1842	1.4229
LSTM	0.184	1.0966
Bidirectional LSTM	0.1841	1.5663

5 Comparative results: SARIMA, hybrid SARIMA–LSTM, and observed data

For comparative purposes, different prediction methods were examined using observed road accident data in the Czech Republic, covering the period from January 2023 to December 2023. The prediction based on the seasonal autoregressive integrated moving average (SARIMA, Eq. 6) can be expressed as the following relation:

$$\begin{aligned}\widehat{Y}_t'' &= 0.9546Y_{t-1}'' - 0.0889Y_{t-2}'' - 0.0110Y_{t-3}'' \\ &+ 0.0326Y_{t-4}'' + 0.0659Y_{t-7}'' + 0.0629Y_{t-8}'' \\ &- 0.0058Y_{t-9}'' - 0.0007Y_{t-10}'' + 0.0021Y_{t-11}'' \\ &+ \widehat{\boldsymbol{e}}_{\boldsymbol{t}} - 1.5821e_{t-1} + 0.586e_{t-2} - 0.9952e_{t-7} \\ &+ 1.574e_{t-8} - 0.5831e_{t-9}\end{aligned} \quad (18)$$

$$\text{where } Y_t'' = Y_t - Y_{t-1} - Y_{t-7} + Y_{t-8}$$

In this formula, the residual term $\widehat{e}_{t+1}$ is unknown. As suggested by previous research [24], this term can be replaced by zero. While such an approximation has little impact on the immediate forecast for t+1, it may introduce additional variability and non-linearity when forecasts are extended over multiple periods.

To enhance predictive accuracy and mitigate the limitations of SARIMA, we employed the Long Short-Term Memory (LSTM) network to model and predict the nonlinear term. The residual forecasts were then integrated into the SARIMA model, yielding a hybrid SARIMA-LSTM approach. This method was applied to the implementation dataset (01.01.2023–31.12.2023) using a one-step-ahead rolling origin strategy with a 7-day sliding window to predict the 8th day.

Table 7 Forecasting performance of SARIMA vs. Hybrid SARIMA-LSTM for the 2023 dataset

Interval	SARIMA RMSE	Hybrid RMSE	Δ RMSE	SARIMA MAPE	Hybrid MAPE	Δ MAPE
30 days	108.0403	108.4973	-0.457	39.4863	39.6166	-0.1303
60 days	77.8906	78.2585	-0.3679	34.0528	34.2049	-0.1521
90 days	78.8907	79.2379	-0.3472	35.8851	36.0004	-0.1154
120 days	80.5057	80.7958	-0.2901	35.8646	35.9426	-0.078
150 days	79.3476	79.6285	-0.2808	33.8005	33.9238	-0.1232
180 days	79.2449	79.5162	-0.2713	33.0669	33.1877	-0.1207
210 days	80.0185	80.3054	-0.2869	34.0163	34.1309	-0.1145
240 days	78.8819	79.1448	-0.263	33.0026	33.0951	-0.0926
270 days	78.9256	79.1756	-0.25	32.7133	32.805	-0.0917
full period	81.5317	81.786	-0.2543	33.2427	33.3327	-0.09

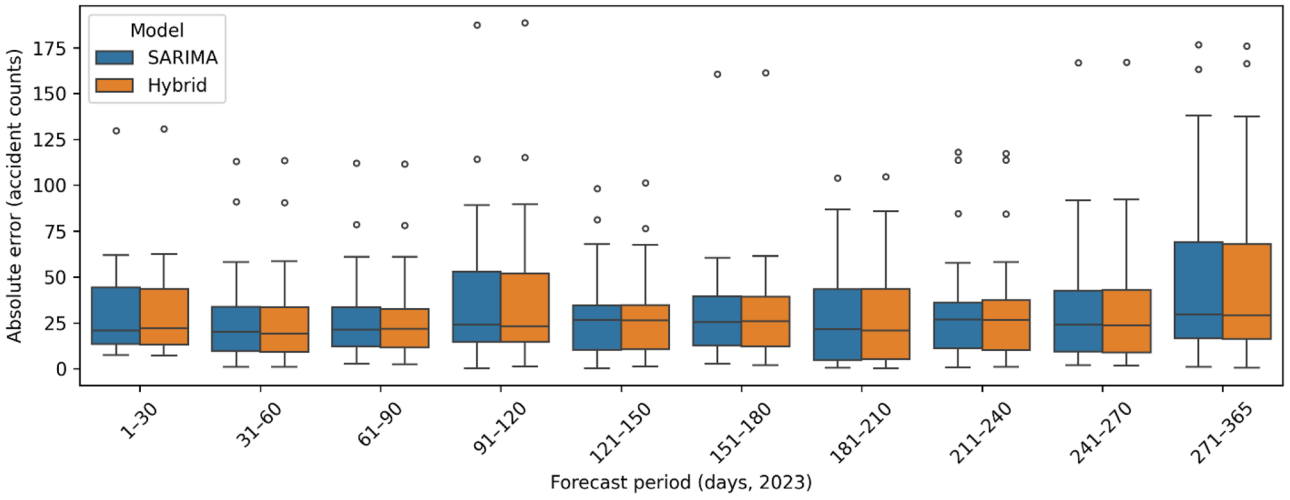


Fig. 6 Distribution of absolute prediction errors for the SARIMA and hybrid SARIMA-LSTM models over the implementation period (01.01.2023–31.12.2023). Boxes represent the interquartile range (IQR), with the horizontal line indicating the median

Table 7 presents a comparative evaluation of the baseline SARIMA model and the hybrid SARIMA-LSTM approach over different forecasting horizons (30–270 days and the full period from 01.01.2023 to 31.12.2023). The performance is measured using RMSE and MAPE, with the difference (Δ) showing the relative change when integrating LSTM-based residual forecasts. Negative Δ values indicate that the hybrid approach slightly increased errors compared to SARIMA, suggesting that for one-step-ahead predictions in this dataset, the added complexity of modelling residual non-linearities did not consistently enhance accuracy.

To complement the numerical comparison in Table 7, Fig. 6 presents the distribution of absolute prediction errors in the 2023 implementation period for both SARIMA and hybrid SARIMA-LSTM models.

The boxplots confirm the consistency of the Table 7 findings: both models exhibit similar error distributions across all periods, with comparable medians, indicating that integrating LSTM-based residual predictions did not meaningfully alter the error structure of the baseline SARIMA forecast at any point during the year.

6 Discussion

This study aimed to evaluate whether nonlinear residual modelling improves short-term accident forecasting when applied under the same seasonal structure and information set as the baseline SARIMA model. To ensure a controlled and fair comparison, both the seasonal autoregressive integrated moving average and the hybrid SARIMA-LSTM framework were trained using identical rolling-origin one-step-ahead forecasting with a 7-day input window corresponding to the weekly seasonal cycle. The analysis, therefore, focuses on isolating the contribution of nonlinear residual learning rather than differences in data availability or model input design.

The experimental results highlight several important insights into the effectiveness of recurrent neural network modelling strategies for capturing non-linearity. First, the comparison of different RNN architectures (Table 6) for choosing to combine with SARIMA demonstrates that memory-enhanced models outperform simple recurrent structures. While the Simple RNN showed the highest error levels across all metrics, the introduction of bidirectionality reduced RMSE and MAPE, suggesting that incorporating both past and future contextual information improves prediction stability. Nevertheless, the unidirectional LSTM achieved the lowest RMSE and MAPE overall, indicating that the LSTM's capacity to capture long-term dependencies outweighed the benefits of bidirectionality in this setting. Interestingly, the bidirectional LSTM did not surpass its unidirectional counterpart, which may reflect overfitting or limited gain from incorporating future information in a one-step-ahead forecasting task.

Second, the hybrid SARIMA-LSTM approach (Table 7) was tested to assess whether modelling residual nonlinearity could improve forecasting accuracy. Across all forecast intervals, from 30 days up to the full period, the differences between the baseline SARIMA and the hybrid model were minimal. ΔRMSE values ranged from – 0.457 to -0.25, and ΔMAPE values were consistently below 0.16, indicating that the hybrid model neither substantially improved nor degraded predictive performance. These results suggest that, for one-step-ahead daily accident forecasting in this dataset, the nonlinear residual component captured by the LSTM provides little additional information beyond the linear SARIMA model. This may be due to the high variability in daily crash counts and

the limited influence of predictable nonlinear trends at the daily level, making the baseline SARIMA model a robust and interpretable choice for operational forecasting. A diagnostic examination of the LSTM-predicted residuals reveals that they were consistently near zero (mean value: $4.67 \times 10 - 5$), indicating that the network effectively produced a null output across the forecasting period. Three candidate explanations are plausible and not mutually exclusive. First, the aggressive differencing pipeline comprising log transformation, seasonal differencing at lag 7, and second-order differencing may have removed all exploitable non-linear structure before the residuals reach the LSTM. Second, the Ljung-Box and Box-Pierce test results ($p = 0.51$ for the selected model) confirm that post-fitting residuals closely approximate white noise with minimal autocorrelation, leaving no systematic temporal pattern for the LSTM to learn. Third, architectural constraints, including a shallow single-layer LSTM, a 40-epoch training budget, and a narrow 7-day input window, may have been insufficient to detect the residual non-linearity that does exist. Distinguishing between these causes definitively would require residual ACF analysis and ablation experiments across network depth and sequence length, both identified as priorities for future work.

Although different combinations of using seasonal autoregression integrated moving average models with an Artificial Neural Network, with high performance metrics: prediction power demand using a hybrid SARIMA and LSTM model [25] where a neural network is responsible for predicting the trend, shows an MAPE of 4.61%, while a SARIMA model with an MAPE of 27.19% can be explained by the input/training data. The data are more stable, and daily power demand does not exhibit significant spikes; fewer unobserved variables influence daily energy consumption, rather than predicting the number of daily crashes.

To contextualise this finding, previous hybrid modelling studies in other domains report substantial improvements [26, 27]; the DWT-SARIMA-LSTM model [25], which combines Seasonal Autoregressive Integrated Moving Average, is used for forecasting wind power. The first step is to decompose the wind power, then predict with SARIMA, and finally use a long short-term memory neural network model. The pre-processing of the data also led to a more stable trend, removing variability and as a result, the hybrid model increases prediction by 13.2%, although very important to mention that the naive model has an R2 value of 86.2% under different weather conditions, which shows a better performance than we have using SARIMA for one-step-ahead prediction of daily crashes. Another relevant application using a hybrid ARIMA-LSTM model [28] predicted vehicle speed with a 15-second horizon in Beijing for the 3rd and 4th Ring roads. The ARIMA models alone achieved low MAPEs of 7.17% and 4.29% for the 3rd and 4th Ring, respectively, while the hybrid ARIMA-LSTM significantly improved performance, with MAPEs of 0.71% and 1.35%, respectively. These studies demonstrate that hybrid models can be highly effective when trends are stable and short-term fluctuations are predictable, in contrast to the more variable daily accident counts in our dataset.

Consistent with prior work demonstrating potential benefits of hybrid modelling [29], the expectation was that residual learning would capture remaining nonlinear dependencies. Our findings suggest otherwise: across all forecast intervals, the hybrid model slightly increased both RMSE and MAPE relative to the baseline SARIMA. The differences, however, were small in magnitude (ΔRMSE and ΔMAPE consistently below 0.5 and 0.2, respectively), indicating that the hybridisation did not cause significant degradation in predictive performance but also failed to yield the expected accuracy improvements.

Taken together, these results suggest that while LSTM networks are promising for capturing temporal dependencies in accident data, their integration with SARIMA residuals requires careful consideration. For one-step-ahead forecasting, the baseline SARIMA model remains robust, and the added complexity of the hybridisation may not provide sufficient benefit. Future research should investigate whether the hybrid approach becomes more advantageous under multi-step-ahead forecasting scenarios, different sequence lengths, or in the presence of stronger non-linear residual structures.

This study is subject to several limitations. First, the model is strictly univariate and does not incorporate exogenous variables known to influence daily accident occurrence, including meteorological conditions, public holiday calendars, and traffic volume dynamics. Such factors can drive sudden spikes in accident frequency that a univariate model cannot anticipate. Incorporating these covariates through a SARIMAX framework or a multivariate LSTM is the most promising immediate extension and the authors' recommended next step. Second, the achieved MAPE exceeding 33% indicates substantial relative error. Comparable univariate daily accident forecasting studies report MAPEs in the range of 20–45% [5, 6, 8], indicating the result is within expected bounds for this problem class; nonetheless, a MAPE below approximately 15% would be required for precise day-level operational deployment. The current model is therefore best interpreted as a directional decision-support signal rather than a precise count predictor. Third, this study does not generate multi-year projections: at daily resolution with a univariate model at 33% MAPE, compounding uncertainty renders a 2–3 year point forecast unreliably wide;

dedicated scenario analysis incorporating exogenous drivers is recommended as a separate future study.

Although the unidirectional LSTM achieved the lowest RMSE and MAPE among the tested models, for the hybrid SARIMA-LSTM, the MAPE exceeding 33% indicates substantial relative error, highlighting that, despite improvements over simpler RNN architectures, the forecasts may have limited practical utility for operational planning, especially when precise day-to-day accident predictions are needed. Users should interpret the predictions cautiously, and future work could explore ways to reduce forecasting error, such as incorporating additional covariates or longer historical sequences.

7 Conclusion

This study evaluated SARIMA and a hybrid SARIMA-LSTM model for short-term road accident forecasting. The results showed that for capturing non-linear trend LSTM networks outperformed simple RNNs, with the unidirectional LSTM achieving the lowest RMSE and MAPE. Bidirectional extensions provided marginal benefits in reducing absolute errors but did not consistently improve overall accuracy.

The hybrid SARIMA-LSTM approach, designed to capture non-linear residual dynamics, produced performance comparable to the baseline SARIMA model but did not deliver consistent gains. The minor increases in RMSE and MAPE suggest that, for one-step-ahead forecasts, SARIMA alone remains highly competitive.

Overall, LSTM-based models offer advantages in capturing temporal dependencies, while SARIMA provides stable and interpretable forecasts. Future work should focus on multi-step-ahead predictions and larger datasets to fully explore the potential of hybrid approaches in modelling accident patterns. This outcome is consistent with the diagnostic finding detailed in the Discussion (Sect. 6): the LSTM-predicted residuals converged near zero throughout the forecasting period, most plausibly because the aggressive differencing pipeline rendered the residuals approximately white noise prior to neural network input.

Therefore, we conclude that while the SARIMA model provides a robust and reliable baseline forecast for road accidents, the implemented hybrid SARIMA-LSTM approach, in this instance, results in a small degraded forecast quality. The pursuit of enhanced accuracy through hybridisation requires meticulous validation of all intermediate outputs to prevent the introduction of new biases. Future work should focus on implementing rigorous prediction-correction techniques for residual forecasts, or on exploring alternative hybrid architectures that can more effectively integrate linear and nonlinear components without compromising the integrity of the trend projection. Ultimately, model simplicity and interpretability, as demonstrated by SARIMA, can often be more valuable than algorithmic complexity if that complexity is not managed with extreme care.

Daily one-step-ahead forecasts can support routine operational planning by identifying expected short-term fluctuations in accident risk. For example, if the model predicts an elevated number of accidents for the following day, regional traffic management centres can proactively adjust patrol deployment schedules, position emergency response teams closer to high-traffic corridors, and increase monitoring during peak-risk periods. Conversely, lower predicted risk allows more efficient allocation of limited personnel and resources. In this way, a reliable univariate daily forecast functions as a decision-support signal for short-term preparedness, enabling authorities to optimise response readiness without requiring complex multivariate data inputs.

Author contributions

I.G. Writing – original draft, Visualization, Methodology, Investigation, Data curation. P.H and E.P -Conceptualization, Supervision, Validation.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study did not involve human participants, animals, or identifiable personal data. The research was based on aggregated and/or anonymized road traffic and accident data. Therefore, ethics approval and informed consent were not required.

Competing interests

The authors declare no competing interests.

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